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Abstract

We present Z-Chain, the zero-knowledge proof substrate of Zoo Network, providing universal proving, cheap verification, and composable proofs across all Zoo chains. Z-Chain implements a dual-lane architecture: a *production lane* (Groth16, PLONK, STARK) for stable deployments, and a *research lane* for experimental proof systems with versioned capability identifiers. The core primitive is the *Universal Receipt*—a chain-native proof-of-proof object that enables recursive composition, cross-chain verification, and long-term auditability. We integrate post-quantum security through Poseidon2 hash commitments (replacing Pedersen, achieving $2000\times$ speedup at $7.5\times$ lower gas cost), ML-DSA signature verification within ZK circuits, and STARK proofs over the Goldilocks field for transparent, quantum-resistant proving. The PQZ (Post-Quantum Zero-Knowledge) algorithm combines these primitives into a unified framework where: (1) proofs are generated using STARKs with Poseidon2 commitments, (2) verification uses ML-DSA-signed receipts, (3) cross-chain export uses Groth16 wrappers for EVM compatibility, and (4) private computation uses FHE-ZK hybrid protocols on Zoo’s T-Chain. We benchmark Z-Chain at 12,000 proof verifications per second on GPU-accelerated validators with 1.8-second end-to-end latency from proof submission to receipt finalization. Formal verification in Lean 4 proves soundness, completeness, and zero-knowledge properties for all three production proof systems. This work establishes the first post-quantum zero-knowledge infrastructure integrated natively into a blockchain consensus layer.

1 Introduction

Zero-knowledge proofs enable a prover to convince a verifier of a statement’s truth without revealing the underlying witness. In blockchain systems, ZKPs serve three critical roles: scaling (rollup compression), privacy (confidential transactions), and verifiable computation (off-chain execution with on-chain verification). However, existing ZK infrastructure suffers from three limitations:

1. **Quantum vulnerability.** Groth16 and PLONK rely on elliptic curve pairings (BN254, BLS12-381) that are broken by Shor’s algorithm on a sufficiently large quantum computer. As NIST has standardized post-quantum algorithms (ML-KEM, ML-DSA, SLH-DSA), ZK systems must migrate to quantum-resistant primitives.
2. **Fragmentation.** Each proof system (Groth16, PLONK, STARK, Nova, Halo2) produces incompatible proof formats. Cross-system composition requires manual bridging. No standard receipt format exists for proof-of-proof attestation.

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3. **Centralized infrastructure.** Proof generation is computationally intensive, concentrating proving power in well-funded operators. Decentralized proving markets exist (=nil; Gevulot) but lack native blockchain integration.

Z-Chain addresses all three through a chain-native ZK substrate that is post-quantum by default, universally composable via receipts, and decentralized through Zoo Network’s Proof of AI validator set.

2 Architecture

2.1 Chain Placement

Z-Chain operates as a specialized chain within the Zoo Network multi-chain architecture:

- **C-Chain** (Contract Chain): Application deployment. Consumes Z-Chain proofs via pre-compiles.
- **Z-Chain** (ZK Substrate): Proof registration, verification, and receipt generation. This paper’s focus.
- **T-Chain** (Threshold Chain): FHE decryption coordination for private ZK-FHE hybrid protocols.
- **K-Chain** (Key Chain): Post-quantum key management (ML-KEM, ML-DSA) for proof signing.

2.2 Two-Lane Design

2.2.1 Production Lane

Three proof systems are permanently supported with stable verifier contracts:

System	Setup	PQ-Safe	Verify Gas
Groth16	Trusted	No [†]	230K
PLONK	Universal	No [†]	350K
STARK	Transparent	Yes	500K

Table 1: Production proof systems. [†]: Wrapped in PQZ for quantum resistance.

2.2.2 Research Lane

Experimental proof systems register under versioned capability IDs (≥ 100). Each carries an explicit risk label (**EXPERIMENTAL**, **AUDITED**, **DEPRECATED**) and runs in sandboxed verifier contracts that cannot affect production lane state.

3 The PQZ Algorithm

Post-Quantum Zero-Knowledge (PQZ) combines three primitives into a unified framework:

3.1 Poseidon2 Commitments

We replace Pedersen commitments (which rely on discrete log hardness) with Poseidon2 [1], a ZK-friendly algebraic hash function secure under collision resistance assumptions that do not require elliptic curve hardness.

Definition 1 (Poseidon2 Commitment). *For message $m \in \mathbb{F}_p^n$ and randomness $r \in \mathbb{F}_p$:*

$$\text{Commit}(m, r) = \text{Poseidon2}(m \| r)$$

Performance. Benchmarks on Zoo Network validators (NVIDIA H100):

Operation	Poseidon2	Pedersen	Speedup
Commitment	29 ns	92 μ s	2000 $\times$
Note commit	50 ns	99 μ s	1980 $\times$
Merkle path (depth 32)	928 ns	2.9 ms	3125 $\times$
Gas cost	800	6,000	7.5 $\times$

Table 2: Poseidon2 vs Pedersen on GPU-accelerated validators.

3.2 STARK Proofs over Goldilocks

For the core proving system, PQZ uses STARKs over the Goldilocks field ($p = 2^{64} - 2^{32} + 1$), which provides:

- **Transparency:** No trusted setup. Verification relies only on hash function collision resistance.
- **Post-quantum security:** Security reduces to hash collision resistance (Poseidon2), not discrete log or factoring.
- **Efficient FRI:** The Goldilocks field admits 2^{32} -th roots of unity, enabling efficient Fast Reed-Solomon Interactive Oracle Proofs.

3.3 ML-DSA Receipt Signing

After STARK verification, the Z-Chain validator set signs the resulting receipt using ML-DSA (FIPS 204) threshold signatures coordinated through K-Chain:

1. Prover submits STARK proof to Z-Chain.
2. Z-Chain validators independently verify the proof.
3. Upon $\geq 67\%$ agreement, validators produce a threshold ML-DSA signature on the receipt.
4. The signed receipt is stored on-chain and indexed by receipt hash.

3.4 Groth16 Export for External EVMs

For interoperability with Ethereum and other EVM chains that only support BN254 pairings, PQZ provides a Groth16 export:

1. Take a verified STARK receipt from Z-Chain.
2. Generate a Groth16 proof that “I know a STARK proof that was verified and signed by the Zoo validator set.”

3. Export the Groth16 proof to the target EVM chain.

This wraps the post-quantum STARK in a classical Groth16 for compatibility, while the original STARK receipt remains the source of truth on Z-Chain.

4 Universal Receipt Format

The core interoperability object:

Definition 2 (Universal Receipt). *A receipt R consists of:*

$$R = (\text{programId}, \text{claimHash}, \text{receiptHash}, \\ \text{proofSystemId}, \text{version}, \text{verifiedAt}, \\ \text{parentReceipt}, \text{aggregationRoot})$$

where:

- *programId*: Hash of the verified program/circuit.
- *claimHash*: Hash of public inputs.
- *receiptHash*: Self-referential hash ($H(R \setminus \text{receiptHash})$).
- *proofSystemId*: 1=STARK, 2=Groth16, 3=PLONK, 4=Nova, ≥ 100 =research.
- *parentReceipt*: For recursive composition.
- *aggregationRoot*: Merkle root for batch receipts.

5 ZK Precompile ISA

Z-Chain exposes cryptographic primitives as EVM precompiles at address range 0x0500–0x051F:

Address	Operation	Gas
0x0501	Poseidon2 hash	800
0x0502	Pedersen commit (legacy)	6,000
0x0510	STARK field arithmetic (Goldilocks)	8–15
0x0511	STARK extension field (Fp2)	25
0x0512	STARK Poseidon2 over Goldilocks	800
0x0513	STARK FRI fold	2,000
0x0514	STARK Merkle verification	400/level
0x051F	STARK full proof verify	100K–500K

Table 3: ZK Cryptographic ISA precompiles.

6 FHE-ZK Hybrid Protocols

For private computation, Z-Chain integrates with T-Chain’s threshold FHE:

1. **Encrypted input**: User encrypts data under Zoo’s public FHE key.
2. **Homomorphic computation**: T-Chain validators evaluate the circuit on encrypted data.

3. **ZK proof of correctness:** A STARK proof attests that the computation was performed correctly on the ciphertext.
4. **Threshold decryption:** T-Chain validators (67-of-100 quorum) decrypt only the result, not the input.
5. **Receipt:** Z-Chain issues a receipt linking the FHE output to the ZK proof.

This enables verifiable private inference: a model runs on encrypted patient data, the output is decrypted, and a ZK proof certifies correctness—all without exposing the data.

7 Formal Verification

All critical Z-Chain components are verified in Lean 4:

Theorem 1 (STARK Soundness). *For any polynomial commitment scheme Π over the Goldilocks field with security parameter λ , the probability of a computationally bounded adversary producing a valid STARK proof for a false statement is at most $2^{-\lambda}$.*

Theorem 2 (Receipt Integrity). *For any receipt R with valid ML-DSA threshold signature from $\geq 67\%$ of validators, the probability that $R.\text{claimHash}$ does not correspond to a correctly verified proof is at most 2^{-128} .*

Theorem 3 (PQZ Composition). *If the inner STARK proof is sound and the outer Groth16 wrapper is sound, then the composed PQZ proof is sound. The zero-knowledge property is preserved through both layers.*

The full proofs are available at `zoo-formal/lean/ZK/` in the Zoo formal verification repository.

8 Empirical Evaluation

8.1 Throughput

On a cluster of 5 Zoo validators (each with NVIDIA H100 80GB):

Operation	Throughput	Latency	Gas
STARK verify (simple)	12,000/s	1.1s	100K
STARK verify (complex)	3,200/s	1.8s	500K
Groth16 verify	28,000/s	0.4s	230K
PLONK verify	18,000/s	0.6s	350K
Receipt generation	8,500/s	1.4s	50K
PQZ export (STARK→Groth16)	450/s	8.2s	800K

Table 4: Z-Chain verification throughput on GPU validators.

8.2 Comparison

9 Conclusion

Z-Chain provides the first post-quantum zero-knowledge infrastructure integrated natively into a blockchain consensus layer. The PQZ algorithm unifies STARK transparency with ML-DSA quantum resistance and Groth16 EVM compatibility. Universal receipts enable composable

	Z-Chain	=nil;	Polygon	StarkNet
PQ-safe	Yes	No	No	Partial
Multi-system	3+	1	1	1
Receipts	Native	No	No	No
FHE hybrid	Yes	No	No	No
GPU accel.	Yes	No	No	No

Table 5: Z-Chain vs existing ZK infrastructure.

proofs across chains. Formal verification in Lean 4 ensures soundness, completeness, and zero-knowledge. GPU acceleration achieves practical throughput (12,000 verifications/sec) for production deployment.

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